# Palash Pawar

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## Education

**The University of Texas at Austin** — Austin, TX
*Bachelor of Science in Statistics & Data Science* — *May 2028*
**Relevant Coursework:** Data Structures & Algorithms, Probability & Statistical Inference, Sports Analytics, Data Visualization, Databases & SQL, Linear Algebra, Machine Learning, Geospatial AI

## Technical Skills

**Languages & Libraries**: Python (pandas, NumPy, scikit-learn, PyTorch, TensorFlow, PySpark), R, SQL
**Data & Cloud Tools**: Databricks, Snowflake, BigQuery, MongoDB, AWS, GCP, Tableau, Power BI, Looker
**Techniques**: Regression, Classification, Forecasting, Causal Inference, LLMs, Computer Vision, A/B Testing

## Experience

**Google x Business of Sports Institute at McCombs** — October 2026 – Present
*Sports Analytics Researcher @ Bevo Lab*
- Building weekly advance scouting reports for Texas Football in BigQuery/Dataform from Catapult/PFF data
- Charting opponent coverage, pressure, and blitz tendencies by down & distance via source-validated LLM analysis

**JPMorganChase** — October 2026
*Data for Good Hackathon | 1st Place Winner (Client: National Basketball Players Association)*
- Modeled 60K NBA games with fixed effects, showing back-to-backs cut win odds ∼6%, validated with XGBoost
- Shipped CourtSide, a D3.js dashboard with an LLM chatbot revealing up to 1.8-win gaps in NBA schedules

**Coinbase** — September 2026
*Insights to Impact Bounty | Top 20 Winner*
- Constructed a signup-to-first-trade funnel (KYC, deposit) unifying 3 event sources with 6 data-quality checks
- Traced a 7-point gap in weekly trader conversion (∼41% vs ∼34%) to a SQL join double-counting trades

**Dell Medical School** — August 2026 – Present
*Undergraduate Research Assistant @ Virostko Lab*
- Developed a multimodal Python pipeline (CGM, wearable, ECG, retinal imaging) for a 100-person diabetes study
- Validated AutoMorph retinal vessel-width measurements as reproducible across cameras (ICC 0.87)
- Cross-validated models: CGM best predicted diabetes stage (AUC 0.77); retina raised systolic-BP $R^2$ $\sim 0 \rightarrow 0.10$

**Databricks** — June 2026 – August 2026
*Data Engineer Intern (Platform Engineering)*
- Architected a Databricks Lakehouse, integrating 3 sources and reducing data latency across 18 production tables
- Designed PySpark ETL pipelines with Delta Lake and Lakeflow orchestration, reducing processing time by 75%

**American Airlines** — August 2025 – January 2026
*Data Scientist Intern*
- Forecasted flight arrival delays with time series models, reducing RMSE by ∼15% vs baseline models
- Engineered temporal features and evaluated seasonal patterns across 2M+ flight records to improve forecasts

**Emerson** — May 2025 – August 2025
*Data Scientist Intern*
- Automated SQL/Azure pipelines to compute, validate, and monitor KPIs across 7+ DeltaV v16.LTS product areas
- Designed Power BI dashboards to visualize release readiness trends and detect high-risk modules in real time

## Projects

**Kalshi Earnings Mentions Bot** | *Claude Code, GCP Compute, SQLite, Kalshi API, SciPy, statsmodels, launchd*
- Backtested a systematic alpha across 5.19M Kalshi earnings-mention contracts (+4.1¢/contract alpha, t = 1.95)
- Deployed a 3-mode trading execution engine with risk-gating, scanning 850 instruments/∼250K API calls per day
- Integrated a Google News sentiment filter, Nasdaq earnings-calendar features, and DuckDB/SQLite tick pipelines

**SPY Volatility Surface & Options Market Maker** | *NumPy, SciPy, pandas, yfinance, pytest, scikit-learn*
- Implemented a SPY options market maker with pricing, Greeks, implied-volatility surfaces, and real-market data
- Developed relative-value signals with inventory-aware quoting, transaction costs, and dynamic delta hedging
- Evaluated with walk-forward testing, achieving a 1.4 Sharpe Ratio and 18% higher risk-adjusted P&L

**Moneyball 2.0: Analyzing Value in the Global Soccer Transfer Market** | *tidyverse, caret, ggplot2*
- Built an end-to-end player valuation pipeline incorporating 30+ advanced performance features
- Analyzed valuation residuals to identify overvalued and undervalued players across leagues and positions

## Activities & Interests

**Activities**: IM Soccer, Texas Dhoom (Bollywood Fusion Dance Team), Stats & Data Science Directed Reading Program
**Interests**: Soccer, Financial/Commodity Markets, Algorithmic Trading, College Football, Traveling, Card Games
**Awards**: 1st Place @ 2026 JPMC Data for Good Hackathon, 3rd Place @ 2024 DECA ICDC, Top 20 in Coinbase Bounty